Characterizing core morphology and scar patterning within a unified spherical harmonic framework

Peiyuan Xiao[a,b,c], Hao Li[a,d,✉], Ben Marwick[c,✉]

Abstract

Lithic cores preserve the most direct evidence of past reduction strategies, yet their analysis still relies largely on typological classifications that are subjective and difficult to reproduce. Recent 3D methods quantify core attributes more precisely, but morphology and scar organization are handled in separate mathematical frameworks, precluding joint analysis of how the two relate. Here we present a unified framework based on spherical harmonics that encodes both core morphology (M-SPHARM) and scar patterning (SP-SPHARM) within a single rotation-invariant, landmark-free space, the latter being the first application of spherical harmonics to scar orientation data. Co-inertia Analysis examines their covariation, and harmonic reconstruction renders statistical axes back into interpretable three-dimensional forms. While the methods are complementary, validation on ideal models and experimental cores shows that the scar pattern descriptor resolves finer distinctions among core types than existing metrics, most clearly the radial subtypes others cannot separate. Exploratory analysis on experimental and archaeological assemblages reveals no detectable covariation between morphology and scar organization. We identify core type as the principal factor structuring variation at Sandinggai (Hunan, China). The methods are released as an open-source R package (spharmlithic) with a fully reproducible workflow. More broadly, the framework supports understanding core variability as a continuous techno-morphological space rather than a set of discrete types.

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Keywords Computational archaeology · Geometric morphometrics · Spherical harmonic · 3D lithic analysis · Core reduction strategy

Introduction

Cores often preserve the most direct evidence of reduction strategies in lithic assemblages. Their analysis can inform not only knapping behavior, including flaking strategies (Delagnes & Meignen, 2006; Inizan & Féblot-Augustins, 1999; Pandey, 2025), reduction intensity (Clarkson, 2013; Lombao, Falcucci, et al., 2023), and knapping skill levels (Cattabriga & Peresani, 2024; Delpiano & Peresani, 2017; Eren et al., 2011; Wilson et al., 2023), but also broader questions of hominin cognition (H. Li et al., 2017; Muller et al., 2022), mobility patterns (Douglass et al., 2018; Lombao, Rabuñal, et al., 2023; Wallace & Shea, 2006), and social learning (Cattabriga et al., 2024; Pargeter et al., 2022; Valletta et al., 2021). Core analysis has traditionally relied on typological approaches, in which cores are classified into categories such as Levallois, discoidal, and laminar cores according to hierarchically organized attributes such as morphology, scar patterning, flaking angles, and platform preparation (Boëda, 1993; Inizan & Féblot-Augustins, 1999). These frameworks, however, are often weakly standardized: different typological systems often employ distinct descriptive vocabularies, and substantial classification inconsistencies arise both between analysts and within the same analyst over time (Pargeter et al., 2023; Ruck et al., 2020; Whittaker et al., 1998). The emerging debate over Levallois technology in China illustrates this problem clearly. Although researchers commonly apply the criteria proposed by Boëda (1995), disagreements persist over whether certain cores from the Shiyu and Guanyindong sites should be classified as Levallois (Carmignani et al., 2024; Hu et al., 2023; F. Li et al., 2019; Yang, Zhang, Yue, Huan, et al., 2024). Efforts to establish unified classificatory systems, such as the Unified Lithic Taxonomy and the Logic Analytical System, have partially reduced inter-observer variability, yet they remain fundamentally reliant on qualitative judgment (Conard et al., 2004; Lombao, Rabuñal, et al., 2023).

Growing demands for precision and reproducibility have driven increasing interest in three-dimensional quantitative approaches to lithic analysis (Grosman et al., 2008; Wyatt-Spratt, 2022). 3D modeling technologies provide substantially higher-resolution data than traditional linear measurements and have been widely applied to flakes (Archer et al., 2021; Bustos-Pérez et al., 2022), retouched tools (Falcucci & Peresani, 2022; Hashemi et al., 2021), and large cutting tools (LCTs) (Herzlinger et al., 2017; H. Li et al., 2021). Studies specifically focused on cores, however, remain limited (Wyatt-Spratt, 2022). Two broad approaches characterize existing work. The first extracts predefined technological variables from 3D models, such as platform angles (Porter et al., 2019), cortex proportions (Lin et al., 2010), reduction intensity (Clarkson, 2013), surface convexity (Muller & Clarkson, 2026), and scar patterning (Clarkson et al., 2006; Lin et al., 2024). The second characterizes overall core morphology, most commonly through landmark-based geometric morphometrics (Hallinan et al., 2026; Hallinan & Cascalheira, 2025; Lycett et al., 2010), though recent studies have also applied spherical harmonic methods (SPHARM) to spheroids and polyhedral cores (Muller et al., 2023; Ye et al., 2026). These two approaches are complementary, and their integration has yielded deeper insights in several studies (Hallinan et al., 2026; Hallinan & Cascalheira, 2025).

These approaches, however, have notable limitations. Landmark-based methods compress continuous morphological variation into a limited number of discrete points, inevitably losing information (Bardua et al., 2019; Goswami et al., 2019; Gunz & Mitteroecker, 2013; Zelditch, 2004). They also rely on homology assumptions: current landmark configurations, which typically place landmarks along core edges and semi-landmarks across surfaces, are mainly applicable to highly standardized systems such as Levallois reduction and difficult to extend to more variable types or to cross-type comparisons (Hallinan et al., 2026; Hallinan & Cascalheira, 2025; Lycett et al., 2010; Lycett & Von Cramon-Taubadel, 2013). Even within Levallois core studies, landmark configurations vary between researchers, making direct comparisons between morphospaces problematic. Spherical harmonic approaches offer a promising alternative. They do not require homology assumptions, capture morphological information more comprehensively, and produce rotationally invariant power spectra that require no prior orientation of the core (Kazhdan et al., 2003; Shen et al., 2009). To date, however, their application has been largely restricted to near-spherical types such as spheroids and polyhedral cores (Muller et al., 2023; Ye et al., 2026), and their potential for more morphologically diverse core types has yet to be tested.

Quantitative approaches to scar pattern analysis are represented by two frameworks: the Scar Pattern Index (SPI), a univariate metric measuring directional concentration through the ratio of resultant to total vector length (Clarkson et al., 2006), and fabric analysis, which uses eigenvalues of orientation tensors to provide a high-level summary of scar patterns (Lin et al., 2024). Both represent important advances, yet both reduce complex scar-vector configurations to a small number of summary parameters that capture only relatively coarse-grained orientation organization rather than the full spatial structural properties of scar patterning itself. This information loss is reflected in their performance: SPI and fabric approaches effectively distinguish broad technological organizations, such as unidirectional or bidirectional removals and Levallois or discoidal cores (radial types), but struggle to differentiate variability within these categories. A method capable of representing the spatial structure of scar organization more completely, rather than reducing it to a few summary statistics, is therefore needed.

The relationship between the way scar organization documents how a core is reduced and the shape it ultimately assumes is an important question in lithic analysis. One influential body of theory holds that knappers actively structured removal sequences to generate and maintain particular core morphologies, most explicitly in prepared core types such as Levallois (Boëda, 1993, 1995; Brantingham & Kuhn, 2001). Other research theorizes that core form is also, and sometimes primarily, shaped by factors largely independent of technological intention, including raw material quality (Dibble, 1991; Kot et al., 2020; Proffitt et al., 2022; Terradillos-Bernal & Rodríguez-Álvarez, 2014), blank geometry (Proffitt et al., 2022; Terradillos-Bernal & Rodríguez-Álvarez, 2014), and reduction intensity (Clarkson, 2013; Lombao et al., 2020). Whether, and to what degree, core morphology reflects deliberate technological design rather than the cumulative by-product of working a given material therefore remains a genuinely open question, with direct bearing on how we infer planning, skill, and cultural transmission from stone tools. Distinguishing between these alternatives requires that technological organization and core morphology be quantified independently and that their covariation be tested directly, something no existing framework achieves within a single, reproducible representation. The problem is compounded by the available tools: scar pattern metrics such as SPI and fabric analysis compress organization into a handful of summary parameters, while landmark-based geometric morphometrics imposes homology assumptions that are tractable only for highly standardized forms and that differ from analyst to analyst. This constrains the cross-type, cross-assemblage, and cross-period comparison on which questions of technological origin and transmission ultimately depend.

Here we develop a spherical harmonic framework that represents core morphology and scar organization within a single mathematical space, recasting core variation from a set of discrete typological categories into a continuous, jointly quantified techno-morphological space. Scar vectors are converted into continuous spherical distributions through von Mises–Fisher kernel density estimation and expanded into rotation-invariant power spectra, placing scar organization in the same framework as morphology, which is described by the same expansion of the core surface. Because the method requires neither homologous landmarks nor manual orientation, its descriptors are directly comparable across core types, raw materials, regions, and time periods. Treating technology and morphology as separable but jointly analyzable dimensions, we use Co-inertia Analysis to test their covariation and exploit the reconstructive capacity of spherical harmonics to translate statistical axes back into interpretable three-dimensional forms. We first validate the framework against digital models of idealized cores and an experimentally knapped set of cores. We then apply the framework in a case study using the core assemblage from Sandinggai, an open air site with Late Pleistocene lithics in central South China. This application serves not as a regional end in itself, but as a worked example of a framework suitable for a variety of broader problems, from the contested status of Levallois in East Asia and the origins of prepared-core technologies to the reconstruction of technological transmission across hominin populations.

Material and Methods

Materials

Three datasets were used in this study, each serving a distinct analytical role. The idealized digital cores dataset was used to evaluate the sensitivity of spherical harmonic methods to standardized scar patterns. The experimentally knapped cores dataset was employed to test the scar pattern analysis framework under more realistic conditions of morphological and technological variability, and to evaluate the integrated statistical framework proposed in this study. The archaeological specimens dataset was then analyzed within this validated framework to demonstrate its practical archaeological applicability.

Idealized digital cores. Eight idealized digital core models, sourced from Lin et al. (2024), were used, comprising cylindrical, conical, discoidal (unifacial and bifacial), Levallois (preferential, convergent, and laminar), biface, and polyhedral types. Each category is associated with distinct and highly recognizable scar organizations, which can be visually examined in the interactive HTML files provided as Online Resource 2.

Experimentally knapped cores. The experimental cores were produced by Chris Clarkson and comprise 58 specimens used in this study: 16 unidirectional, 6 bidirectional, 21 Levallois (10 preferential, 5 convergent, 2 Nubian, 2 recurrent, and 2 laminar), 5 discoidal, and 10 multiplatform cores. These specimens are curated at the School of Social Sciences, University of Queensland. The cores were digitized by Sam Lin using a Polyga HDI Compact C210 structured-light scanner with FlexiScan3D software (Lin et al., 2024).

Archaeological specimens. The archaeological specimens derive from Sandinggai (hereafter SDG) in Hunan Province, China, previously reported by H. Li et al. (2022). The SDG sequence comprises three cultural layers (L2, L3, and L4), dated by optically stimulated luminescence (OSL) to approximately 100–20 ka. The lithic assemblage is characterized by two operational sequences: (1) a small-flake system primarily based on chert, and (2) a large-flake and large cutting tool (LCT) system primarily based on quartz sandstone (H. Li et al., 2022).

From the previously reported SDG core assemblage, 51 complete and well-preserved specimens retaining clear technological information were selected for analysis, after excluding incomplete, heavily abraded, and test cores. These derive from Layers 2, 3, and 4; however, because only two cores originate from Layer 2, our statistical analyses were restricted to Layers 3 and 4. Core classifications follow the original paper, and details of core types and raw materials are provided in Table 1. Archaeological cores were digitized using an EinScan Pro XS handheld 3D scanner (SHINING 3D) with a turntable and fixed camera stand. Model alignment and export were completed using the accompanying EinScan-Pro software.

Table 1 Specimen inventory of the analysed Sandinggai (SDG) core assemblage, listed by core type, raw material, and stratigraphic layer. Counts include the analysed cores from Layers 3 and 4; the two Layer 2 cores were excluded from the statistical analyses because of their insufficient sample size. Coretype classifications follow the original study.

Core type	Chert	Sandstone	Layer 3	Layer 4	Total
Unifacial unidirection	7	0	4	3	7
Unifacial centripetal	12	0	5	7	12
Bifacial adjacent	4	4	4	4	8
Bifacial independent	3	0	1	2	3
Bifacial centripetal	1	0	0	1	1
Multifacial	6	7	3	10	13
Core on flake	4	1	1	4	5
Total	37	12	18	31	49

For both experimental and archaeological datasets, post-processing was conducted in Geomagic Wrap 2021, including the filling of minor holes and the correction of mesh defects such as non-manifold edges and self-intersections.

Data Extraction

Scar orientations were recorded following the protocol established by Lin et al. (2024). Only scars larger than 2 mm were included. Each scar was recorded as a vector defined by the three-dimensional coordinates of its start and end points, using Geomagic Wrap 2021 for data acquisition. A best-fit plane was also calculated for each core model, and the coordinates of the plane centroid and normal vector were exported for model alignment.

Two alignment procedures were employed to eliminate initial phase differences among core models. The first follows Lin et al. (2024), aligning models based on their best-fit planes by rotating the plane normal to coincide with the Z-axis and translating the starting point of the longest scar to the global origin. The second procedure is proposed here for the first time and relies exclusively on the spatial organization of scar data: (1) unit vectors are calculated for all scars and subjected to singular value decomposition (SVD); (2) the third singular vector, representing the direction least correlated with overall scar orientation structure, is selected as the alignment-plane normal and rotated to the Z-axis; (3) the global centroid of all scar coordinates is translated to the origin; and (4) the first singular vector, representing the principal direction of scar organization, is aligned with the X-axis. The original vectors and the vectors aligned under both procedures were then used to evaluate the rotational invariance of the scar pattern metrics described below. To allow direct visual inspection of the alignment outcomes, interactive HTML visualizations of each model under both procedures are provided as Online Resource 2.

Spherical Harmonic Analysis

Spherical harmonic analysis represents the three-dimensional extension of elliptic Fourier analysis (Harper et al., 2022; Shen et al., 2009). The method characterizes surfaces of closed genus-zero objects by mapping each vertex onto the unit sphere through equivalent coordinate functions and representing the surface as a weighted sum of spherical harmonic basis functions (Brechbühler et al., 1995; Shen et al., 2009). The resulting shape descriptors are invariant to rotation, translation, and scaling, and can be used to reconstruct the original three-dimensional morphology directly (Kazhdan et al., 2003). SPHARM was originally developed for morphometric analysis of brain structures in medical imaging and has since been applied across multiple disciplines (Gerig et al., 2001; Medyukhina et al., 2020; Radvilaitė et al., 2016). In lithic studies, Sholts et al. (2017) used spherical harmonics to quantify asymmetry in Clovis points, while Muller et al. (2023) and Ye et al. (2026) applied the method to spheroid morphology. Beyond morphological characterization, mathematical relationships between spherical harmonic representations and directional data have also been established (Lee et al., 2022). In the present study, we exploit this relationship by applying SPHARM to scar orientation data for the first time, as shown in Fig. 1.

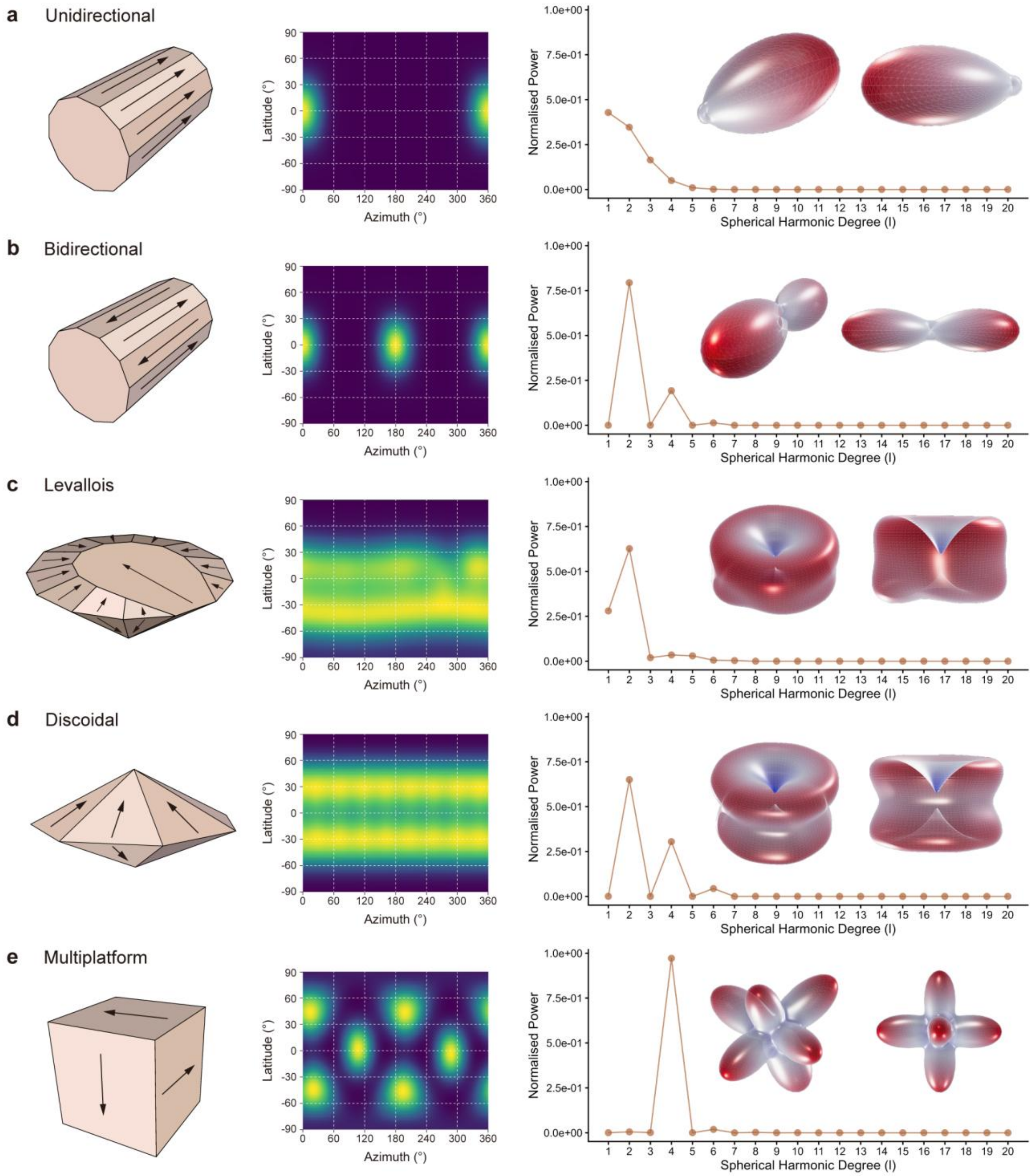

Fig. 1 Application of spherical harmonic analysis to scar orientation data, demonstrated on five idealized core types: **(a)** unidirectional, **(b)** bidirectional, **(c)** Levallois, **(d)** discoidal, and **(e)** multiplatform. For each type, the idealized 3D core model and its scar pattern are shown on the left, converted into a continuous spherical density using von Mises–Fisher kernel density estimation in the middle, and represented by a rotation-invariant power spectrum. The scar pattern reconstructed from the harmonic coefficients is shown on the right

Our analytical workflow for scar pattern analysis proceeds as follows. Discrete scar vectors are first transformed into a continuous density distribution on the sphere using von Mises–Fisher spherical kernel density estimation (VMF-sKDE) (Alonso-Pena et al., 2024; Bai et al., 1988):

$$f(\mathbf{g}) = \frac{1}{n}\sum_{i=1}^{n} \exp\left(\kappa\, \mathbf{g} \cdot \mathbf{x}_i\right)$$

where $\mathbf{g}$ is a unit vector on the spherical grid, $\mathbf{x}_i$ is the unit vector representing the orientation of the i-th scar, and $\boldsymbol{\kappa}$ is the concentration parameter of the von Mises–Fisher kernel, related to the bandwidth $\mathbf{h}$ by:

$$\kappa = \frac{1}{h^2}$$

A bandwidth of 0.35 ($\boldsymbol{\kappa} \approx 8.16$) was selected to balance directional resolution against oversmoothing. This choice was validated by reconstructing the original distributions, in which distinct scar patterns remained clearly identifiable.

The KDE distributions were interpolated onto a 64×128 Driscoll–Healy grid and subjected to spherical harmonic expansion. Each distribution $f(\theta, \phi)$ is represented as a weighted combination of spherical harmonic basis functions $Y_l^m(\theta, \phi)$:

$$f(\theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \hat{f}\,(l, m)\, Y_l^m(\theta, \phi)$$

where the coefficients $\hat{f}(l, m)$ quantify the contribution of each basis function, and **l** and **m** denote harmonic degree and order, respectively. A truncation threshold of $l_{\max} = 20$ was adopted in our study, consistent with previous studies demonstrating that this degree sufficiently captures detailed artifact morphological variation (Ye et al., 2026). The resulting coefficients retain phase information and can be used to reconstruct original three-dimensional forms, but phase sensitivity may introduce noise in comparative analyses. Accordingly, scar pattern spherical harmonic (hereafter SP-SPHARM) power spectra were also calculated by summing squared coefficients within each harmonic degree and normalizing to unit total, providing a phase-independent, rotationally invariant representation of scar pattern structure.

For core morphological analysis, we followed the procedures established by Ye et al. (2026). Three-dimensional models (.stl/.ply) were downsampled to 20,000 faces, and Laplacian smoothing was applied to reduce surface noise and regularize vertex distribution. Centroid normalization and principal-axis alignment were then performed to position all vertices within a common coordinate system, which was transformed into spherical coordinates for harmonic expansion. Spherical harmonic coefficients and morphological spherical harmonic (hereafter M-SPHARM) power spectra were calculated at $l_{\max} = 20$, as for the scar pattern analysis. The complete workflow is illustrated in Fig. 2.

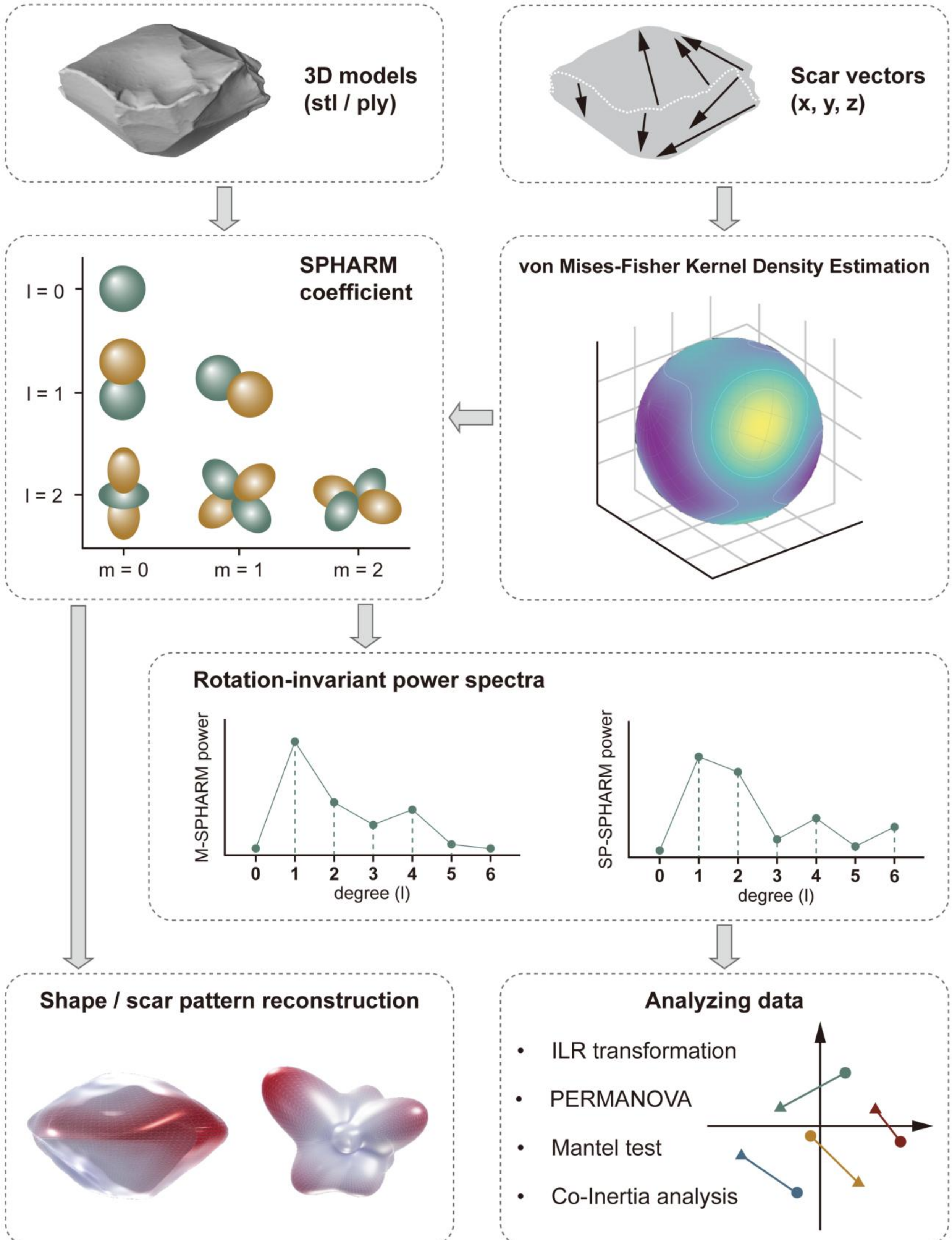

Fig. 2 Overview of the unified spherical harmonic workflow for jointly describing core morphology and scar organization

Although coefficients were computed to $l_{\max} = 20$ for faithful reconstruction, retaining all 20 degrees for statistical comparison would add dimensionality and, for scar patterns, noise. We therefore screened degrees prior to all statistical analyses using two diagnostics: the across-specimen coefficient of variation (CV), indexing the reliability of each degree's between-specimen signal, and cumulative power, indexing the share of total variance each degree carries. The two spectra behaved differently and were truncated on different grounds. For SP-SPHARM, power was effectively exhausted by the sixth degree (cumulative energy ≈ 99.9%), and higher degrees were noise-dominated (CV > 100% from **l** ≈ 9); we therefore retained **l** = 1–6. For M-SPHARM, between-specimen signal remained reliable across all degrees (CV < 90% throughout), but variance was strongly concentrated in the low degrees, with ~98% of cumulative power captured by the eighth degree; we retained **l** = 1–8, since the reliable but very low-energy higher degrees (together under 2% of power) would inflate dimensionality and dilute multivariate comparisons without materially improving discrimination. This contrast confirms that scar pattern spectra are markedly lower-dimensional than morphological spectra (Fig. 3).

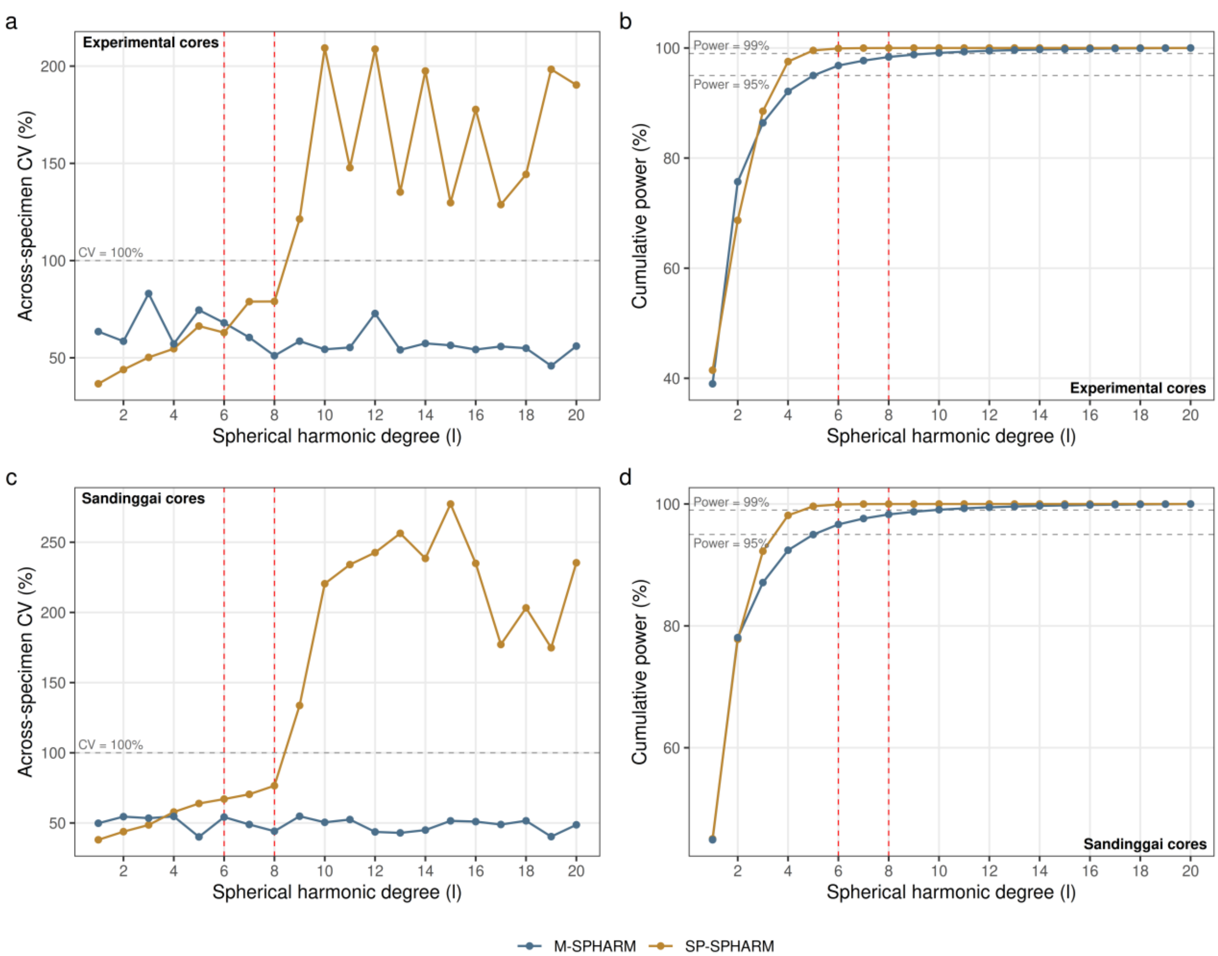

Fig. 3 Per-degree diagnostics used to select the spherical-harmonic truncation degree, shown for the experimentally knapped cores and the Sandinggai assemblage. **(a, c)** Across-specimen coefficient of variation (CV) by harmonic degree l; **(b, d)** cumulative power by harmonic degree l. In each panel the morphology (M-SPHARM) and scar pattern (SP-SPHARM) power spectra are overlaid. Dashed lines mark the 100% CV reference in (a, c) and the 95% and 99% cumulative-power references in

(b, d)

Data Analysis

Rotation Invariance Assessment

Comparing specimens with these metrics presupposes that each captures the scar pattern itself rather than the arbitrary orientation in which a model happens to be digitized. We therefore first evaluated whether each metric is independent of model orientation, that is, whether its value is stable across the original orientation and the two alignment procedures described above. Rotational invariance of three scar pattern metrics (SPI, fabric Elongation and Isotropy, and SP-SPHARM power spectra) was assessed using the Bland–Altman approach. This method evaluates agreement between paired measurements by plotting their difference against their mean (Bland & Altman, 1986, 1999). The mean difference represents systematic bias, and the 95% limits of agreement (LoA) define the range within which 95% of differences are expected to fall; narrow LoA with no systematic trends indicate high agreement (Bland & Altman, 1999). Each metric was computed under three coordinate conditions: (1) original scar-vector data, (2) morphological-plane alignment, and (3) technological-plane alignment. Pairwise Bland–Altman comparisons among these conditions were conducted, with high agreement interpreted as evidence of rotational invariance.

Method Comparison

For the idealized digital cores, where only one model exists per type, discriminatory power was assessed using pairwise distances in feature space, with greater distances indicating stronger inter-type separation; because the SP-SPHARM power spectra are compositional, we used standardized Euclidean distances on ILR coordinates (Aitchison, 1982), whereas SPI and fabric used standardized Euclidean distances on the raw features. Results were visualized using heatmaps. For the experimentally knapped cores, a significance-testing framework was employed. SPI, as a univariate metric, was evaluated using the Kruskal–Wallis test (Kruskal & Wallis, 1952). Fabric metrics and SP-SPHARM descriptors were analyzed using PERMANOVA, which tests overall structural differences between groups, complemented by PERMDISP to assess whether differences in within-group dispersion contribute to the observed separation (Anderson, 2001, 2006). Significant PERMDISP results warrant cautious interpretation of PERMANOVA outcomes (Anderson, 2006, 2017). All permutation procedures used 9,999 permutations. Where overall tests were significant, *post hoc* pairwise comparisons were performed with Holm–Bonferroni correction (Holm, 1979).

Techno-morphological statistical analytical framework

To investigate techno–morphology relationships, we employed a two-step analytical framework: first testing whether overall covariation exists between M-SPHARM and SP-SPHARM power spectra, then exploring interaction patterns through joint dimensional reduction.

Isometric log-ratio transformation. Spherical harmonic power spectra are compositional data: the total power is constrained to a fixed sum, so variation in one harmonic degree necessarily induces spurious covariance among others (Aitchison, 1982; Greenacre, 2021). To eliminate this closure effect, Isometric log-ratio transformation (ILR) was applied to all power spectrum data, projecting them into Euclidean space through log-ratio calculations among harmonic degrees (Egozcue et al., 2003). All subsequent statistical analyses were conducted on ILR-transformed data. For the comparison analyses in which SP-SPHARM is placed alongside the non-compositional SPI and fabric metrics, the ILR coordinates were additionally standardized so that distances are expressed on a common scale across methods.

Mantel test and RV coefficient. Overall covariation was evaluated using two complementary approaches. The Mantel test assesses correlation between two distance matrices via Spearman's rank correlation coefficient, with significance determined by 9,999 permutations (Mantel, 1967). The RV coefficient quantifies the overall degree of association between two multivariate datasets, ranging from 0 (no association) to 1 (identical structure), with significance likewise assessed by permutation (Escoufier, 1973; Robert & Escoufier, 1976). The Mantel test emphasizes pairwise distance relationships, whereas the RV coefficient focuses on multivariate covariance structure; their combined use therefore provides a more robust assessment of dataset correspondence.

Co-inertia analysis. Co-inertia Analysis (CoIA) is a multivariate method for evaluating structural relationships between two datasets. Its core principle is to identify directions of variation along which the two datasets are maximally concordant while maximizing their covariance (Dolédec & Chessel, 1994; Dray et al., 2003). PCA is first performed separately on each dataset, after which projections are used to identify correspondences among samples and variables across the two data spaces (Dolédec & Chessel, 1994; Dray et al., 2003). CoIA is particularly well suited for comparing datasets that differ in physical meaning but share identical sampling units, making it especially appropriate for jointly analyzing M-SPHARM and SP-SPHARM power spectra, which describe different aspects of core variability across the same set of specimens.

The CoIA ordination axes represent the principal directions of shared variation between the two datasets. Each sample, in our case each core, is represented by two points, one from the M-SPHARM analysis, and one from the SP-SPHARM analysis, connected by a line segment. Line length reflects the degree of discrepancy between a sample's position in the two data spaces: shorter lines indicate stronger correspondence between its morphological and scar pattern signatures, whereas longer lines indicate that the two dimensions characterize that specimen differently (Dolédec & Chessel, 1994; Dray et al., 2003). Line direction reflects the trend of covariation for that sample within the shared space. If line directions are broadly consistent across a group of samples, this suggests a stable and internally coherent covariation structure. Dispersed directions suggest weaker structural correspondence or the presence of localized variation. Combined with the reconstructive capacity of spherical harmonics, this framework allows the morphological and scar pattern characteristics associated with each end of the CoIA axes to be directly visualized as three-dimensional forms, thereby reconnecting statistical patterns to concrete archaeological interpretation.

To further characterize sample correspondences, we employed Kruskal–Wallis tests to compare line lengths among groups, assessing whether the degree of technology–morphology coupling differs across core types (Kruskal & Wallis, 1952). Rayleigh tests were applied to line directions to evaluate whether orientations exhibit significant directional clustering (Mardia, 1988). Because line directions in CoIA represent axial correspondences rather than unidirectional vectors, they were treated as axial data, with directions differing by 180° considered equivalent (Arnold & SenGupta, 2006; Mark, 1973). The combined analysis of line lengths and directions

334 enables simultaneous evaluation of coupling strength and structural consistency from both
335 distance-based and directional perspectives. The whole workflow is shown in Fig. 4.

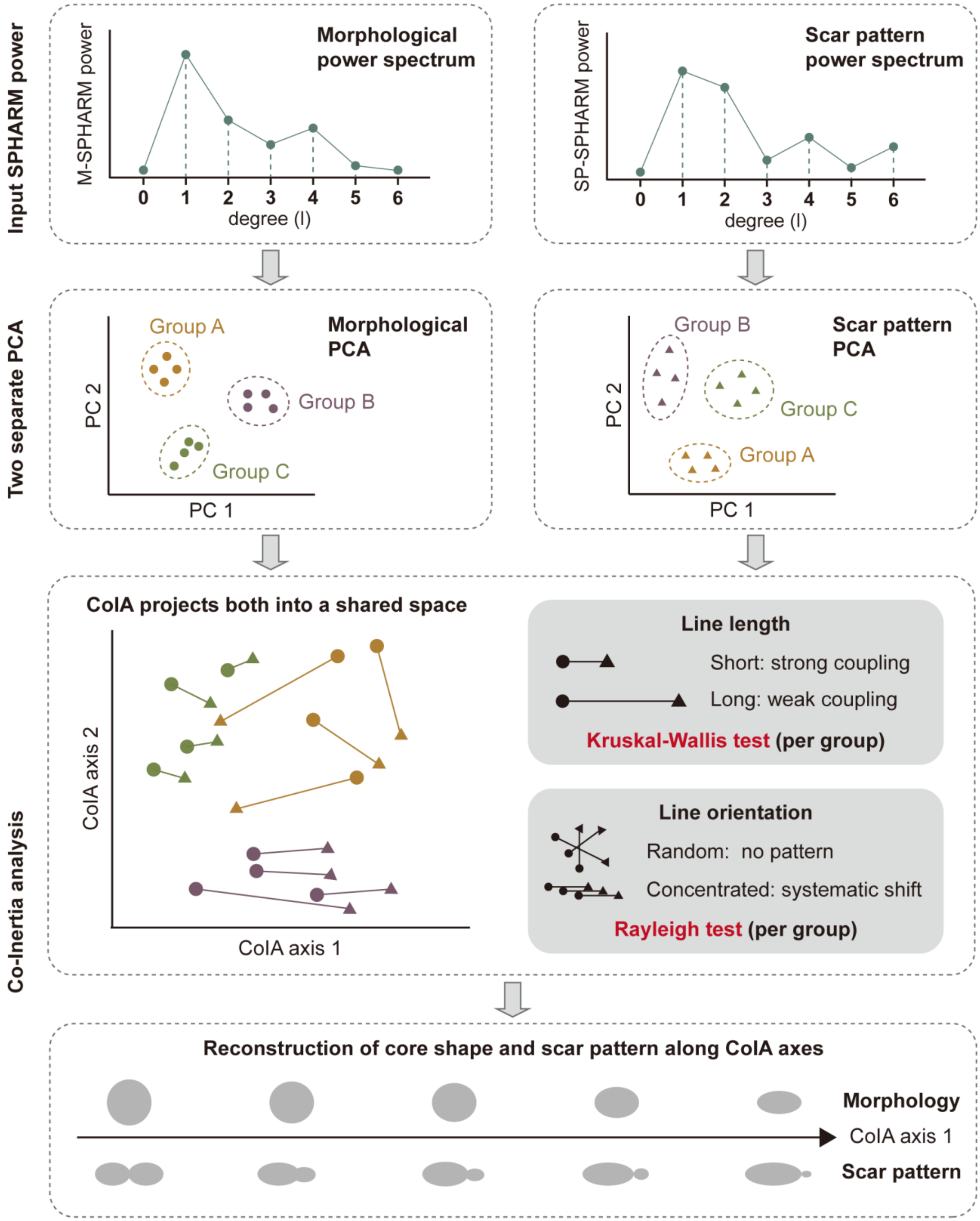

Fig. 4 Workflow of the Co-inertia Analysis (CoIA) framework for jointly analysing morphology and scar pattern

Reconstruction along CoIA axes. To visualize the coupled techno–morphological variation captured by CoIA, we developed a continuous reconstruction framework operating directly in spherical harmonic coefficient space. Spherical harmonic coefficients of individual specimens were projected onto the CoIA axes, and at evenly spaced positions along each axis, reconstructed forms were generated by Gaussian kernel-weighted averaging of neighboring specimens' coefficients (Hutton et al., 2003). The weighted coefficients were then transformed back into three-dimensional meshes (M-SPHARM) or spherical density surfaces (SP-SPHARM) through inverse spherical harmonic reconstruction. The resulting forms represent localized structural tendencies along covariance gradients rather than static group averages, enabling gradual transitions in morphology and scar organization to be visualized as continuous trajectories within the shared CoIA space.

Sensitivity analysis. Because the proposed descriptors involve several analyst-defined computational parameters, sensitivity analysis provides a necessary check on whether the reported patterns reflect empirical structure in the data rather than idiosyncratic parameter choices (Hamby, 1994; Saltelli, 2002). To confirm that our conclusions do not depend on specific parameter choices, we conducted parameter sensitivity analysis varying the four principal free parameters of the two descriptors: the von Mises–Fisher kernel bandwidth and the minimum scar size threshold for SP-SPHARM, and the mesh decimation target and number of Laplacian smoothing iterations for M-SPHARM. Full procedures and complete results are reported in the Online Resource 1; the outcomes relevant to each analysis are noted in the Results and Discussion below.

Power analysis. To evaluate whether non-significant association results could be attributed to limited sample size, we assessed the range of covariation that the present sample design could reasonably detect. Rather than calculating observed power from the realized effect, which is not informative for interpreting a completed test (Hoenig & Heisey, 2001), we evaluated what size of association the present sample design could reasonably detect (Lakens, 2022). To do this, we combined three checks: a bias-corrected RV coefficient, bootstrap estimates of the plausible upper range of the association, and simulations testing whether Mantel and RV tests would detect known levels of covariation at the realized sample sizes (Rohlf & Corti, 2000). The procedure and full results are reported in Online Resource 1.

Reproducibility and open source materials

Computational reproducibility remains an ongoing challenge in archaeological science, yet it is essential for the credibility and cumulative progress of quantitative research (Marwick, 2017; Marwick et al., 2017). To support reproducibility and re-use, all materials necessary to reproduce the analyses presented here are provided in version-controlled online repository at OSF (https://doi.org/10.17605/OSF.IO/D82C5). These materials include the complete R and Python scripts used for all analyses and visualizations, and the raw datasets underlying all reported results. The repositories are organized following established best practices for research compendia, with a logical folder structure and relative file paths that allow all code to be executed without manual modification. All figures, tables, and statistical results presented in this paper can therefore be independently reproduced from these materials.

In addition, we provide an open-source R package (spharmlithic) that implements the essential analytical methods developed in this study, including model alignment, SPI and fabric computation, spherical harmonic analysis of both scar patterns and core morphology, and three-dimensional visual reconstruction (Xiao & Marwick, 2026). This package is designed as a general-purpose tool to enable other researchers to apply these methods to their own datasets.

Results

Validity Analysis

Rotational Invariance of Scar Pattern Analysis Measures

To assess whether scar pattern metrics require coordinate alignment prior to analysis, rotational invariance was evaluated using the Bland–Altman approach. SPI and fabric metrics showed near-perfect agreement across all three coordinate conditions (original, morphological-plane aligned, and technological-plane aligned), with biases and 95% LoA on the degree of 10^{-16}, well within floating-point numerical precision (Goldberg, 1991; Higham, 2002). Both measures are therefore effectively rotationally invariant, and coordinate alignment is not required prior to their computation (Fig. 5).

SP-SPHARM descriptors exhibited small systematic biases in low-degree power ($\mathbf{l}$ = 1–2; 95% LoA = ±0.02), which diminished rapidly with increasing degree; no substantive differences were observed for $\mathbf{l} \geq 3$ (Fig. 5). Random rotational perturbations around the Z-axis produced differences within machine precision, confirming that directional uncertainty within the coordinate frame has no effect on the power spectrum given a fixed Z-axis.

The low-degree biases most likely originate from upstream computational steps such as KDE interpolation and coordinate mapping rather than from the power spectrum formulation itself. Crucially, a consistent alignment method applied uniformly across all specimens ensures that this systematic error does not alter relative differences among them. All subsequent analyses therefore employ the technological fitted plane for alignment.

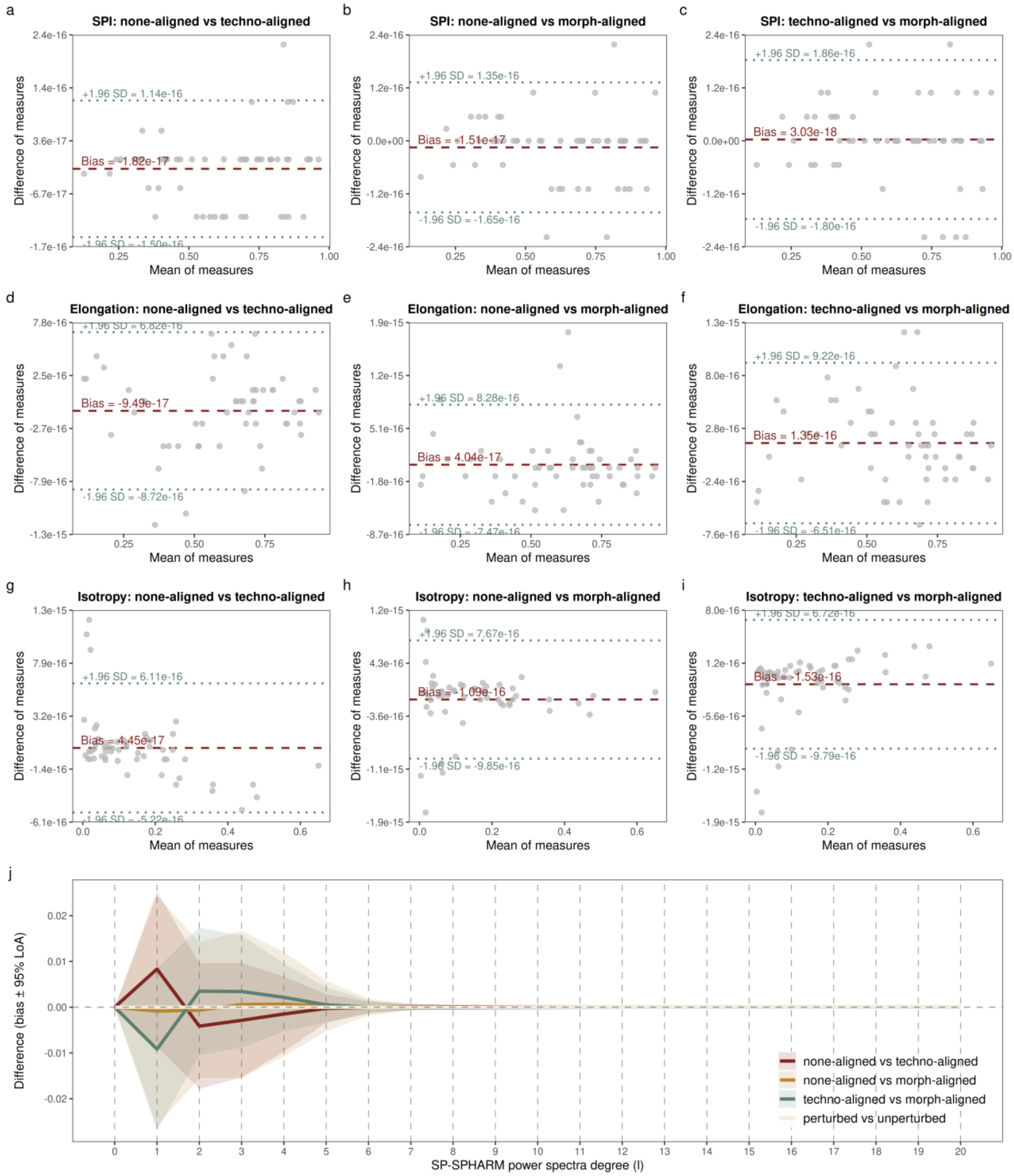

Fig. 5 Bland-Altman analysis of rotational invariance. **(a-c)** SPI across three coordinate conditions. **(d-f)** Fabric elongation ratio across three coordinate conditions. **(g-i)** Fabric isotropy ratio across three coordinate conditions. **(j)** SP-SPHARM power spectra bias across three coordinate conditions and random Z-axis perturbation

Validity of the SPHARM Method for Scar Pattern Analysis

Test of idealized models. Inter-type separation increased progressively from SPI to fabric metrics to SP-SPHARM descriptors, as revealed by the pairwise distance heatmaps (Fig. 6), where SP-SPHARM distances are standardized Euclidean distances on ILR coordinates and SPI and fabric distances are standardized Euclidean distances on the raw features. Mean distances confirmed this ordering: SP-SPHARM yielded the greatest overall separation ($\bar{d}_{\mathrm{E,std}}$ = 1.94), exceeding fabric metrics ($\bar{d}_{\mathrm{E,std}}$ = 1.66) and SPI ($\bar{d}_{\mathrm{E,std}}$ = 1.12).

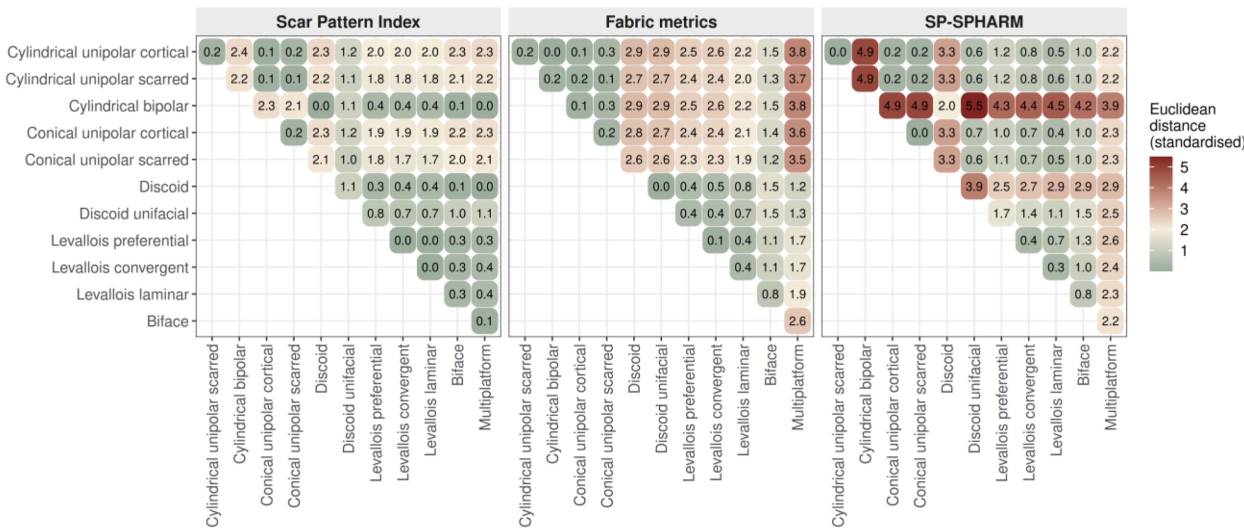

Fig. 6 Discriminatory power of the three scar pattern analytical methods on the idealized digital cores, shown as heatmaps of pairwise standardized distances between core types for SPI, fabric metrics, and SP-SPHARM power spectra. For SP-SPHARM the distances are standardized Euclidean distances on ILR coordinates; for SPI and fabric they are standardized Euclidean distances on the raw features. Larger distances indicate stronger separation

SPI produced moderate distances between unidirectional types (cylindrical, conical) and radial or centripetal types (discoidal, Levallois), but near-zero distances among Levallois subtypes (preferential, convergent, and laminar) and between discoidal variants, indicating that it primarily captures the degree of directional concentration while remaining insensitive to finer structural differences. Fabric metrics substantially improved separation between unidirectional and radial categories but, like SPI, failed to distinguish scar patterns sharing parallel but opposing orientations. Distances between unipolar and bipolar cylindrical types remained near zero. SP-SPHARM produced the highest pairwise distances across nearly all comparisons. Notably, it achieved clear separation between discoidal and Levallois types and between bifacial and unifacial discoidal patterns, distinctions that neither SPI nor fabric metrics resolved. Multiplatform cores were most strongly differentiated from all other types under SP-SPHARM. However, distances between types sharing bipolar symmetry in the power spectrum domain, such as bifacial discoidal and bipolar cylindrical patterns, remained comparatively modest, foreshadowing a limitation also observed in the experimentally knapped core analysis below.

Test of experimentally knapped cores. To evaluate the sensitivity of each method on empirical data, we computed all three measures for experimentally knapped cores and conducted statistical comparisons by core type. All three methods detected significant inter-type differences (Fig. 7), but they resolved different sets of distinctions. SP-SPHARM descriptors separated the broadest set of type pairs, including several that neither summary metric could. Fabric metrics distinguished the majority of type pairs but failed to differentiate unidirectional from bidirectional cores. SPI was the most limited, detecting differences only where unidirectional cores were involved.

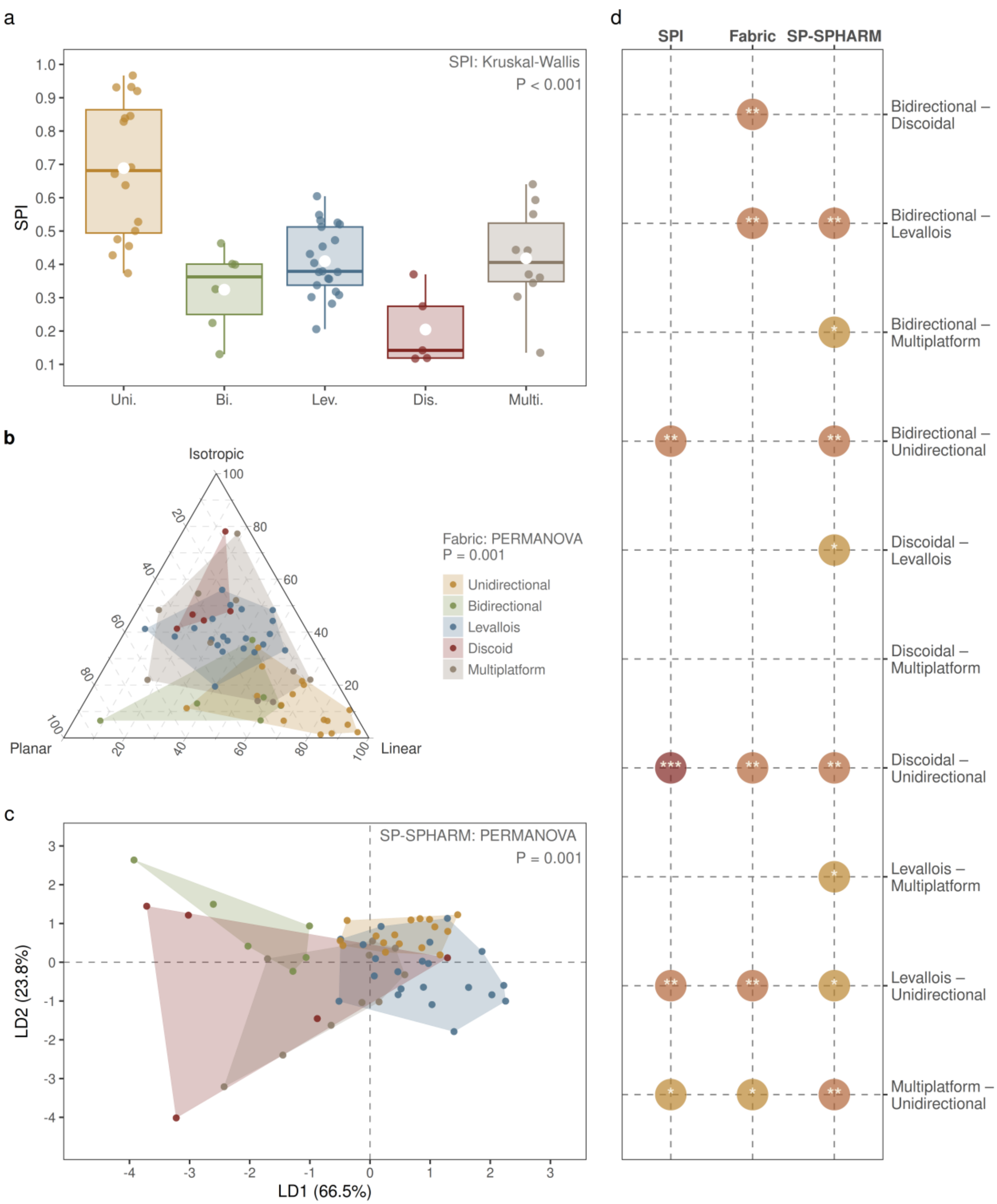

Fig. 7 Comparison of the three scar pattern analytical methods on the experimentally knapped cores, grouped by core type. **(a)** Box-and-whisker plot of SPI distributions by core type with Kruskal–Wallis testing result. **(b)** Fabric metrics shown as a Benn ternary diagram of elongation, planar, and isotropy components with PERMANOVA result. **(c)** SP-SPHARM descriptors projected onto the first two linear discriminant axes with PERMANOVA result. **(d)** Core type pairs significantly separated by each method after Holm–Bonferroni correction; each marker denotes a pair distinguished by the corresponding method. The three methods detect significant inter-type differences but resolve different sets of pairwise distinctions

These results suggest that the three methods are complementary. SP-SPHARM descriptors capture the overall structure of scar patterning for most pairwise comparisons, though separation is reduced for patterns with bipolar symmetry, such as discoidal-bidirectional pairs (p_{adj} = 0.3807). Fabric metrics effectively distinguishes differences between unidirectional and bidirectional cores from Levallois and discoidal cores, but cannot differentiate unidirectional from bidirectional cores (p_{adj} = 0.0476), nor Levallois from discoidal cores (p_{adj} = 0.0325). SPI is most sensitive to the unidirectional type but shows limited discriminative capacity for other types. Notably, although SP-SPHARM descriptors failed to separate discoidal from bidirectional cores due to their similarity structures in SPHARM power spectrum (Fig. 6), this distinction is successfully recovered by fabric metrics (p_{adj} = 0.0132). PERMDISP on the SP-SPHARM descriptors was significant (p = 0.003), driven mainly by the greater within-group dispersion of discoidal (and, more weakly, Levallois) cores relative to unidirectional cores. We therefore interpret the PERMANOVA cautiously, noting that the two non-significant pairs (discoidal–bidirectional and discoidal–multiplatform) both involve the discoidal group, which combines high dispersion with a small sample (n = 5).

Joint Analysis of Scar Patterns and Core Morphology in Experimental Knapping Specimens

PERMANOVA tests using M-SPHARM descriptors showed core type to account for a small but statistically significant fraction of morphological variation (R^2 = 0.123, pseudo-F = 1.864, p = 0.025), with no individual type pair resolved after Holm-Bonferroni correction. Despite this weak structuring of morphology by core type, the global Mantel test likewise revealed no significant correlation between M-SPHARM and SP-SPHARM descriptor sets (Mantel's r = -0.061, p = 0.796), and the RV coefficient from CoIA did not reach significance (RV = 0.101, p = 0.12). Both association measures were negligible in magnitude, indicating that core morphology and scar organization vary largely independently at the assemblage level.

Because the global tests above provide no evidence of covariation, the CoIA axes are interpreted here descriptively, as the directions of maximal shared variation extracted by the method, rather than as a statistically supported covariance structure. The first two CoIA axes account for 85.6% of total covariation (Axis 1 = 65.5%, Axis 2 = 20.1%; Fig. 8). Along Axis 1, M-SPHARM reconstructions shift from rounded, equidimensional forms to progressively flattened profiles, while SP-SPHARM reconstructions transition from concentrated, directionally organized density distributions to more dispersed, multidirectional patterns characteristic of centripetal or multiplatform scar organizations. Axis 2 captures a secondary morphological gradient from laterally convex, disk-shaped cross-sections to more uniform, flattened elliptical forms. On the scar pattern side, both axes capture broadly similar transitions in scar organization, reflecting a shift from ordered, directionally structured arrangements, such as unidirectional and bidirectional patterns, toward more dispersed, multidirectional configurations typical of discoidal and multiplatform cores (Fig. 8).

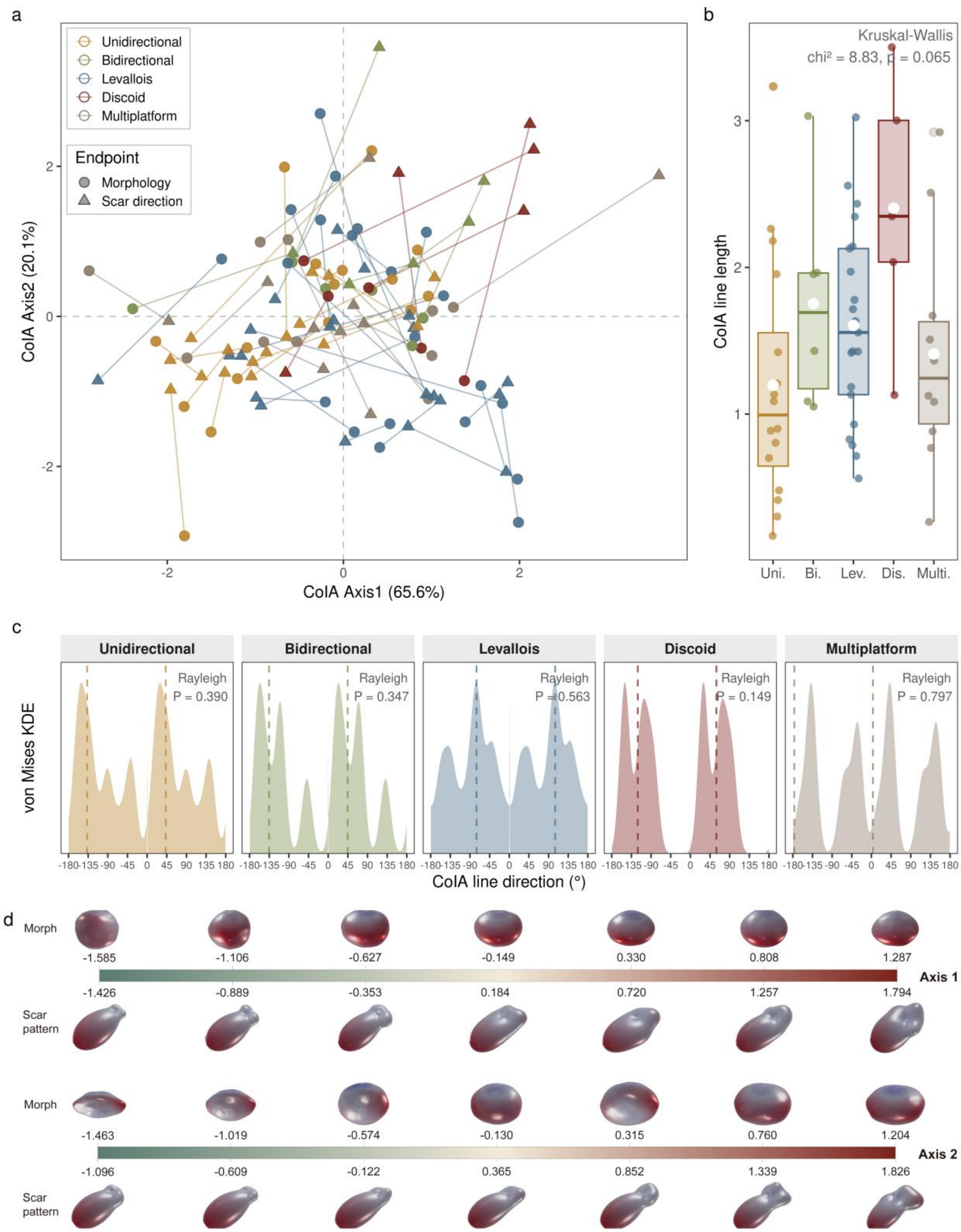

Fig. 8 Co-inertia analysis (CoIA) of M-SPHARM and SP-SPHARM descriptors for the experimentally knapped cores. **(a)** CoIA ordination on the first two axes; each core is shown as a morphology point (circle) and a scar pattern point (triangle) joined by a line, whose length and direction summarize the morphology–scar pattern correspondence for that specimen. **(b)** Line lengths compared among core types using Kruskal–Wallis test. **(c)** Rose diagrams of line directions by core type with Rayleigh testing results. **(d)** Morphology (top) and scar pattern (bottom) reconstructed along the CoIA axes by inverse spherical harmonic reconstruction

Kruskal–Wallis tests revealed no significant difference in line lengths across core types (Fig. 8), indicating that the degree of morphology–scar correspondence does not differ detectably across reduction strategies. Overall, the experimental assemblage shows no detectable covariation between core morphology and scar pattern: both the global Mantel test and the RV coefficient are negligible. At the within-type level, no core type showed significant directional clustering of CoIA line directions (all Rayleigh $p > 0.05$). This broader pattern suggests that morphology and scar organization vary primarily along relatively independent axes; under the current sample conditions, no statistically meaningful techno-morphological coupling trajectory can be detected.

Archaeological Case Study: Core Analysis from Sandinggai

As with the experimental cores, the global Mantel test revealed no significant correlation between M-SPHARM and SP-SPHARM descriptors for the SDG assemblage (Mantel's r = 0.012, p = 0.427). Cross-frequency Mantel tests remained non-significant after Holm–Bonferroni correction, confirming that no covariation is detectable in any frequency band. The RV coefficient likewise did not reach significance (RV = 0.091, p = 0.313).

As in the experimental assemblage, the non-significant global tests mean that the CoIA axes below are read descriptively rather than as evidence of a real covariance structure. The first two CoIA axes account for 90.6% of total covariation (Axis 1 = 57.6%, Axis 2 = 33.0%; Fig. 9). Along Axis 1, M-SPHARM reconstructions transition from flattened, dorsoventrally compressed forms to more rounded, equidimensional profiles, while SP-SPHARM reconstructions shift from concentrated, unidirectionally organized density distributions to more symmetrically opposed, bidirectional patterns. Axis 2 captures additional variation in both dimensions. On the scar pattern side, it distinguishes unidirectional and bidirectional organizations from more centripetal, dispersed configurations. On the morphology side, it primarily reflects mid-frequency shape variation associated with three-fold and four-fold rotational symmetry, likely corresponding to differences in core cross-sectional geometry (Fig. 9).

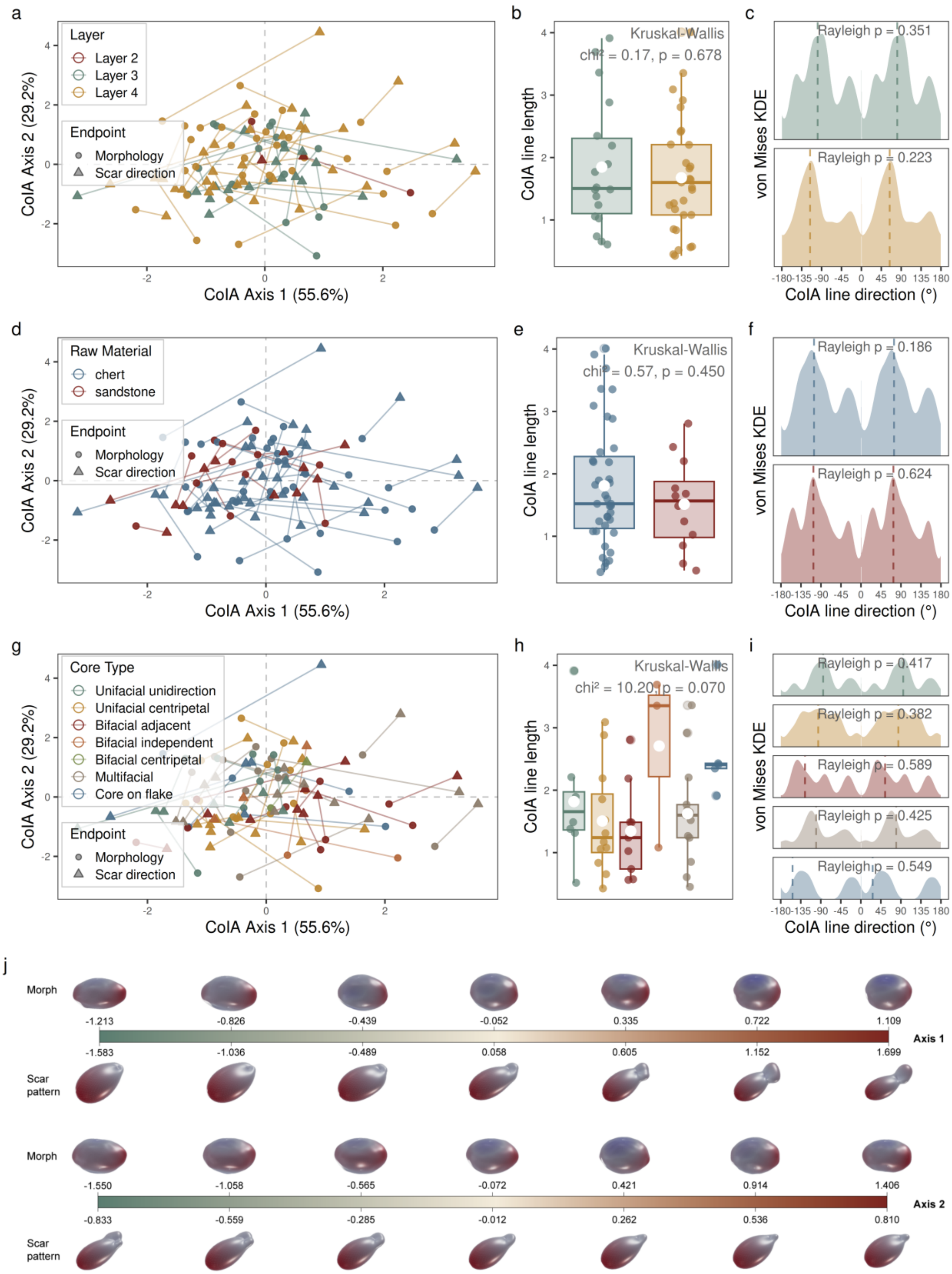
a
b
c
Layer
Layer 2
Layer 3
Layer 4
Endpoint
Morphology
Scar direction
ColA Axis 2 (29.2%)
ColA Axis 1 (55.6%)
Kruskal-Wallis
chi² = 0.17, p = 0.678
ColA line length
Rayleigh p = 0.351
Rayleigh p = 0.223
von Mises KDE
ColA line direction (°)
d
e
f
Raw Material
chert
sandstone
Endpoint
Morphology
Scar direction
Kruskal-Wallis
chi² = 0.57, p = 0.450
Rayleigh p = 0.186
Rayleigh p = 0.624
g
h
i
Core Type
Unifacial unidirection
Unifacial centripetal
Bifacial adjacent
Bifacial independent
Bifacial centripetal
Multifacial
Core on flake
Endpoint
Morphology
Scar direction
Kruskal-Wallis
chi² = 10.20, p = 0.070
Rayleigh p = 0.417
Rayleigh p = 0.382
Rayleigh p = 0.589
Rayleigh p = 0.425
Rayleigh p = 0.549
j
Morph
Scar pattern
Axis 1
Axis 2
-1.213 -0.826 -0.439 -0.052 0.335 0.722 1.109
-1.583 -1.036 -0.489 0.058 0.605 1.152 1.699
-1.550 -1.058 -0.565 -0.072 0.421 0.914 1.406
-0.833 -0.559 -0.285 -0.012 0.262 0.536 0.810

Fig. 9 Co-inertia analysis (CoIA) of M-SPHARM and SP-SPHARM descriptors for the Sandinggai core assemblage, with specimens grouped by stratigraphic layer **(a–c)**, raw material **(d–f)**, and core type **(g–i)**. In each grouping, the CoIA ordination (left) shows every core as a morphology point (circle) and a scar pattern point (triangle) joined by a line; the centre panel compares line lengths among groups (Kruskal–Wallis test) and the right panel shows rose diagrams of line directions (Rayleigh test). **(j)** Morphology (top) and scar pattern (bottom) reconstructed along the CoIA axes by inverse spherical harmonic reconstruction

Line lengths showed no significant differences among layers (χ^2 = 0.17, df = 1, p = 0.678), raw materials (χ^2 = 0.57, df = 1, p = 0.45), or core types (χ^2 = 10.2, df = 5, p = 0.07; Fig. 9), indicating that the degree of morphology–scar correspondence does not differ detectably across any grouping condition. Rayleigh tests on line direction revealed no significant directional concentration in any layer, raw material, or core type (all p > 0.05). The absence of any directionally structured group indicates that the decoupling between morphology and scar organization holds consistently down to the within-type level, with no core type showing a structured techno-morphological trajectory.

PERMANOVA identified core type as the dominant factor structuring both dimensions, with significant and comparable effects on morphological and scar pattern variation, whereas raw material and layer each accounted for substantially less variation (Table 2). Raw material explained only a small, non-significant fraction of morphological variation and had a negligible effect on scar pattern, while layer effects were negligible for both dimensions, consistent with continuity in knapping behavior across the stratigraphic sequence.

Table 2 PERMANOVA results for the Sandinggai core assemblage, Layers 3 and 4 only. Each variable (core type, raw material, and layer) was tested in a separate PERMANOVA using 9,999 permutations, for both the morphology (M-SPHARM) and scar pattern (SP-SPHARM) descriptors. R^2, proportion of variance explained; pseudo-F, PERMANOVA test statistic; p, permutation-based p-value.

Term	df	M-R^2	M-pseudo-F	M-p	SP-R^2	SP-pseudo-F	SP-p
Core type	5	0.188	2.030	0.010	0.168	1.780	0.040
Raw material	1	0.041	2.080	0.080	0.018	0.905	0.435
Layer	1	0.018	0.842	0.491	0.020	0.948	0.415

Two patterns emerge from these effect sizes. First, core type explained roughly four to five times more variation than either raw material or layer in both dimensions, identifying it as the principal axis of structured variability at SDG. Second, although core type significantly structured both morphology and scar organization, the two dimensions showed no detectable pairwise covariation (Mantel's r = 0.012; RV = 0.091): type-level differences in form and in scar organization are not aligned along a shared axis, each responding to core type but largely independently of the other. The somewhat larger effect of raw material on morphology (R^2 = 0.041) than on scar organization (R^2 = 0.018) is consistent with material properties influencing form more than reduction organization, but both effects are small and this tendency should not be over-interpreted at the present sample size.

Discussion

The principal contribution of this study is conceptual as much as methodological. By embedding scar organization and core morphology within a single rotation-invariant, landmark-free mathematical framework, it treats lithic core variability as a continuous techno-morphological space rather than a collection of discrete types. This framework enables scar patterning and form to be quantified separately and their relationship assessed directly. In this section we first evaluate the robustness of the approach across diverse reduction systems, then examine what the decoupling of scar organization and morphology reveals about knapping behaviour, and finally we consider its implications for long-standing cross-regional questions. By providing a homology-free, period-agnostic descriptor of core variation, our framework offers analytical possibilities beyond those available to conventional landmark-based approaches.

The validity of SP-SPHARM for characterizing scar organization is supported at multiple levels. Visual inspection of reconstructed idealized models, experimentally knapped cores, and the SDG specimens reveals clear and interpretable differences in scar pattern, and statistically, PERMANOVA confirms that SP-SPHARM discriminates most scar-pattern categories. A reasonable concern is whether this discrimination is driven disproportionately by small, less reliably recorded scars. A scar-size sensitivity analysis indicates that it is not (Online Resource 1: Table S2): as scars at or below 5 mm and then 10 mm are progressively excluded, the overall PERMANOVA by core type remains significant at every threshold (R^2 = 0.282–0.322, all $p <$ 0.05), and pairwise resolution stays high throughout, separating eight of ten pairs at the >2 mm baseline, seven at the >5 mm cutoff, and eight at the >10 mm cutoff. The core type signal is therefore carried primarily by larger, more reliably recorded scars and is robust to excluding the smallest removals, confirming that the separation reflects a genuine technological signal rather than noise carried by small scars. The non-significant distinction between discoidal and multiplatform cores is particularly informative. It reflects the high internal variability of the multiplatform category, within which certain specimens exhibit scar patterns resembling those of discoidal cores. Importantly, this overlap does not imply that these specimens should be reclassified (Lin et al., 2024). Rather, it shows that scar organization does not map one-to-one onto conventional typological categories but encodes technological variation that those categories — which bundle morphology and scar pattern into a single label — cannot isolate. This reinforces the value of quantifying scar organization as a dimension of core variability distinct from, and complementary to, morphology.

Compared with existing quantitative approaches to scar pattern analysis, SP-SPHARM demonstrates clear advantages, although it does not constitute a complete replacement for either the SPI or the fabric method. SPI is fundamentally a measure of directional concentration. It quantifies the degree to which scar orientations converge toward a single resultant direction and is therefore most sensitive to unidirectional patterns, while performing poorly in discriminating among radial or centripetal scar organizations (Clarkson et al., 2006). The fabric method adds a second analytical dimension by characterizing variation in orientations perpendicular to the best-fit core plane, thereby enabling separation between bidirectional and radial core types that SPI cannot achieve (Lin et al., 2024). However, because fabric metrics are derived from orientation tensor eigenvalues, they summarize axial orientation structure rather than vector polarity. In the eigenvalue method for axial data, reversing the sense of an observation does not change the orientation matrix (Mark, 1973), while fabric-shape indices such as elongation and isotropy describe whether orientations form linear, planar, or isotropic configurations (Benn, 1994; McPherron, 2018). Consequently, geometrically parallel scar pattern that differ mainly in technological polarity, such as unidirectional and bidirectional arrangements, may produce similar fabric configurations, as also suggested by Lin et al. (2024). In contrast, SP-SPHARM encodes the spatial structure of scar organization more completely, rather than compressing it into summary parameters. This is what allows it to separate radial subtypes such as Levallois and discoidal cores, a distinction that neither SPI nor fabric metrics achieved. However, SP-SPHARM reconstructions represent continuous density distributions that cannot be straightforwardly mapped onto discrete scar pattern categories. By contrast, SPI and fabric indices retain stronger direct physical interpretability. In practice, an effective workflow is to first assess overall group differences using SP-SPHARM, and then integrate SPI and fabric analyses to refine archaeological interpretation.

At the level of practical implementation, the three methods also differ in their sensitivity to coordinate alignment. SPI and fabric metrics are strictly rotationally invariant, requiring no alignment prior to computation. SP-SPHARM power spectra show small systematic biases in low-degree terms attributable to upstream computational steps, necessitating a consistent alignment procedure across all specimens. This additional requirement is a practical cost, but one that is offset by the finer structural distinctions SP-SPHARM resolves. More broadly, all three quantitative approaches share an important advantage over traditional scar pattern classification: they replace subjective categorical judgments with continuous numerical descriptors, thereby reducing inter-observer inconsistency and enabling reproducible, statistically testable comparisons across assemblages.

Beyond method validation, this study integrates M-SPHARM and SP-SPHARM power spectra into a CoIA framework to directly test whether core morphology and scar patterning covary at the assemblage level. This addresses a long-standing gap in core analysis: the lack of a quantitative method capable of simultaneously characterizing both dimensions within a unified analytical framework. Results from the experimentally knapped cores indicate that neither the global Mantel test nor the RV coefficient reaches statistical significance, and the CoIA line-direction analyses reveal no structured techno-morphological covariation within core types. Together, these results point to a general absence of detectable covariation between morphology and scar organization: at least at this sample size, similar reduction strategies can produce a range of core morphologies, and conversely, similar morphological outcomes can arise from different technological organizations.

This finding has both theoretical and methodological implications. In terms of theories of core exploitation, the absence of detectable covariation suggests that core morphology and reduction strategy respond to different constraints and should not be treated as interchangeable proxies for one another in archaeological interpretation. This does not imply that conventional core typologies are invalid. Rather, it suggests that their coherence depends on dimensions beyond global morphology and global scar organization, including locally expressed technological features such as flaking angle, platform preparation, and the organization of specific flaking surfaces. The relationship between technological intention and core form may therefore be better addressed through these more localized and technologically diagnostic variables than through overall core morphology alone. Methodologically, it validates the analytical approach adopted here, in which morphology and scar pattern are characterized as separate yet complementary dimensions rather than collapsed into a single typological classification. The CoIA framework, combined with the reconstructive capacity of spherical harmonics, further enables this variation to be visualized as continuous trajectories along covariance gradients rather than as discrete typological categories, moving beyond static classification toward a dynamic representation of techno-morphological variability. As an exploratory tool, CoIA remains informative even under decoupling, since it characterizes the overall structure of variation within an assemblage. Where sample sizes permit, a more powerful application would be to run it separately within each core type, capturing the within-type spectrum of techno-morphological variation that our present per-type samples are too small to resolve. This decoupling is robust to analytical choices: across all vMF kernel bandwidths, scar size thresholds, and mesh preprocessing settings tested, the morphology–scar pattern association remained negligible in magnitude (Mantel's $r \approx 0$; $RV \approx$ 0.06–0.12 across both assemblages; Online Resource 1). RV crossed the 0.05 threshold only under the two most extreme settings (an over-smoothed kernel and an unsmoothed mesh), where it remained negligible and was uncorroborated by the Mantel test. A complementary power analysis further suggests that this pattern is unlikely to reflect limited sample size alone, since moderate associations would probably have been detected (Online Resource 1).

Our application of the framework to the SDG assemblage provides a direct test of its performance on archaeological materials while yielding new behavioral insights. Previous studies proposed that lithic technology at SDG remained broadly continuous across layers, with two principal operational sequences: a small-flake system based on chert and a large-flake and LCT system based on quartz sandstone (H. Li et al., 2022). However, whether these two raw material systems involve distinct reduction strategies at a finer behavioral level has not been fully explored. The present results provide quantitative support for technological continuity: no significant differences in either core morphology or scar pattern were detected between Layers 4 and 3, confirming that reduction behavior remained stable across the stratigraphic sequence.

At the same time, the framework clarifies the relative roles of the factors structuring core variation at SDG. Core type was the dominant influence on both morphology and scar organization, explaining several times more variation than raw material or stratigraphic layer. Raw material had only a small, non-significant effect on morphology and a negligible effect on scar organization, while layer effects were negligible throughout. This pattern indicates that knapping behavior at SDG was structured primarily by the kind of core being produced rather than by raw material or by time, consistent with a stable technological tradition. The capacity to attribute variation to these competing factors, and to show that morphology and scar organization respond to them largely independently, would be difficult to achieve without the separate quantification of both dimensions afforded by the framework. More broadly, wherever assemblages combine multiple raw materials, the same approach can investigate theories about whether material differences are expressed in form, in reduction organization, or in both. This is a question of broad relevance to the role of raw material in the lithic record. We found this decoupling held consistently across the SDG assemblage: no core type, raw material, or stratigraphic layer exhibited a distinct structured techno-morphological trajectory, reinforcing the conclusion that core morphology and reduction organization vary independently at SDG.

At a broader perspective, the framework proposed here offers potential analytical pathways for addressing several unresolved issues in current studies. For example, the presence of Levallois technology in China has been a matter of persistent contention, with ongoing disagreements over whether specific cores from sites such as Guanyindong and Shiyu fulfil Levallois criteria (Carmignani et al., 2024; Hu et al., 2019, 2023; F. Li et al., 2019; Yang, Zhang, Yue, Wood, et al., 2024; Yang, Zhang, Yue, Huan, et al., 2024). The SPHARM framework, by providing continuous quantitative descriptors independent of subjective typological boundaries, may offer additional analytical perspectives for such debates. Similarly, the evolutionary relationship between Levallois technology and Acheulean industries remains a central question in Lower–Middle Paleolithic transition research globally. Competing hypotheses including independent invention, derivation from handaxe traditions, and development from earlier prepared-core technologies (Adler et al., 2014; Agam et al., 2022; Moncel et al., 2020; Rosenberg-Yefet et al., 2021; Shipton et al., 2013). Recent landmark-based geometric morphometric studies have compared Levallois cores, simple prepared cores, and handaxes, suggesting multiple regional trajectories of origin (Gill et al., 2025). However, landmark configurations may vary between researchers, limiting cross-study integration and large-scale comparative potential (Bardua et al., 2019; Collyer et al., 2020; Webster & Sheets, 2010). Because SPHARM does not rely on homologous landmarks and produces standardized spectral descriptors, it offers a potential solution to this limitation. Although the current application focuses on cores and does not yet include large cutting tools, the method is in principle applicable to such type. To support adoption of this framework, we provide an open-source R package (spharmlithic) that implements the main analytical methods developed in this study, including model alignment, SPI and fabric computation, spherical harmonic analysis of both scar patterns and core morphology, and three-dimensional visual reconstruction (Xiao & Marwick, 2026). This well-documented package enables students and researchers to easily and freely apply the methods developed here to their own datasets without requiring specialist programming expertise.

More broadly, the approach has potential for integration with emerging analytical paradigms. Recent work has incorporated elliptic Fourier series into phylogenetic analyses, though these applications have been restricted to two-dimensional morphologies of flaking products (Matzig et al., 2024). As a three-dimensional generalization of Fourier methods, spherical harmonic analysis could be extended to phylogenetic frameworks. The scar pattern component developed here is particularly relevant in this regard, as it provides a direct line of evidence for reconstructing reduction strategies. Future integration with phylogenetic modeling may offer new opportunities to investigate technological transmission and processes of social learning among hominin groups.

Several limitations should be acknowledged. First, sample sizes for certain core types, particularly bidirectional and discoidal cores in the experimental assemblage, are small, and the current results should therefore be interpreted as exploratory rather than definitive. Second, the experimental cores were produced by a single knapper, which may not fully capture the range of inter-individual variability expected in archaeological assemblages. Finally, the present study operates at the core level, characterizing overall morphology and scar organization as aggregate properties. Finer-grained technological attributes, such as platform preparation, flaking surface maintaining details, are not captured by the current framework. Future study should combine quantitative analyses across multiple scales, progressing from macroscopic to microscopic levels, in order to comprehensively characterize lithic variability and more clearly elucidate hominin technological strategies.

Conclusion

This study introduces a spherical harmonic framework for the integrated three-dimensional analysis of lithic cores, extending SPHARM beyond near-spherical artifacts to a broader range of morphologically diverse core types. The framework characterizes core morphology (M-SPHARM) and scar pattern (SP-SPHARM) within the same mathematical representation, using von Mises–Fisher kernel density estimation to convert discrete scar vectors into continuous spherical distributions amenable to harmonic expansion.

Validation against digital idealized cores and experimentally knapped cores demonstrates that SP-SPHARM resolves finer distinctions than existing approaches, most notably separating radial core subtypes such as Levallois and discoidal that neither SPI nor fabric metrics can resolve. The three methods nonetheless capture different aspects of scar organization and are best used in combination: SP-SPHARM for overall structural comparison, SPI and fabric metrics for interpretively grounded refinement.

Building on these descriptors, we further introduce a Co-inertia Analysis framework as a companion approach for jointly examining technological and morphological variation. Applied to experimental and archaeological assemblages, this exploratory framework reveals no detectable covariation between core morphology and scar organization at the assemblage level, a decoupling that held broadly across core types, raw materials, and stratigraphic layers. These findings should be regarded as preliminary, but they illustrate the analytical possibilities opened up by treating morphology and scar pattern as separate yet jointly analyzable dimensions.

Finally, in support of broader adoption of our framework, we offer spharmlithic, an open-source R package including walk-through documentation. This conveniently implements the essential analytical methods developed in this study, including model alignment, SPI and fabric computation, SPHARM analysis of both morphology and scar patterns, and three-dimensional visual reconstruction. The package, together with the version-controlled repository containing the complete analysis scripts and data for this paper, enables both independent reproduction of the reported results and application of the methods to new datasets.

Future work should focus on expanding sample sizes and assemblage diversity, and integrating SPHARM with phylogenetic and other emerging quantitative paradigms to investigate broader questions of the technological behavior of past humans.

Statements and Declarations

Funding

This work was financially supported by the Archaeological Talent Promotion Program of China (No. 2025-198, to H. L.), the National Natural Science Foundation of China (Nos. U2574202 and 42471180, to H. L.), and the China Scholarship Council Program for International Students (CSC No. 202504910326, to P. X.).

Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Author Contributions

P. X.: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. H. L.: Funding acquisition, Resources, Supervision, Writing – review & editing. B. M.: Conceptualization, Methodology, Software, Supervision, Validation, Writing – review & editing. All authors reviewed and approved the final manuscript.

Data Availability

All data and code required to reproduce this study, including the 3D core models, scar orientation and observational data, and the complete R and Python analysis code, are archived in the project's OSF repository (https://doi.org/10.17605/OSF.IO/D82C5). The repository includes a README file that describes its structure in detail and provides step-by-step instructions for reproducing the analyses.

Acknowledgements

We are grateful to Chris Clarkson of Griffith University and Sam Lin of the University of Wollongong for generously sharing the idealized digital core dataset and the experimentally knapped core dataset, and for permitting us to make this high-quality dataset publicly available. We also thank Yiyuan Li of the Hunan Provincial Institute of Cultural Relics and Archaeology for providing access to the Sandinggai specimens for this study.

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